

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459479.6782 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1603 +/- 0.0043
Transit depth (Rp/Rs)^2	2.57 +/- 0.14 [%]
Orbital Inclination (inc)	85.04 +/- 0.39
Airmass coefficient 1 (a1)	1.175 +/- 0.017
Airmass coefficient 2 (a2)	-0.136 +/- 0.023
Scatter in the residuals of the lightcurve fit is	1.41 %
Best Comparison Star	None
Optimal Aperture	4.69
Optimal Annulus	15.67
Transit Duration (day)	0.0792 +/- 0.0037

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.1603 ± 0.0043 vs 0.1326 (deviation 5.74σ)
Duration fit vs published	0.0792 d vs 0.0737 d (deviation 1.31σ)
Mid-time O-C	-4.6 min
Depth SNR	33.2
Residual scatter	1.41 %

Light curve

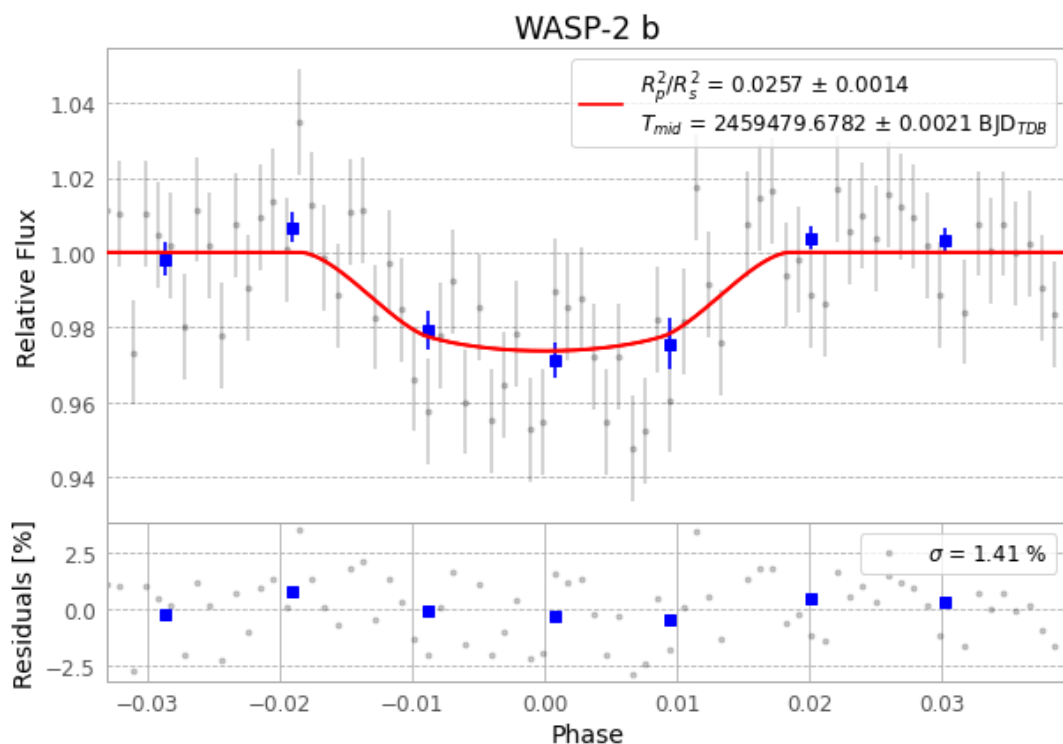


Figure 1 — FinalLightCurve_WASP-2 b_2021-09-21.png (SHA-256 0211e954c5563a87...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [368, 89], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e684c94cae55b0d6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

76 FITS frames (50 MB), WASP-2210922022712.FITS → WASP-2210922061213.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-210922.log (d327a5428fdb...), FinalLightCurve_WASP-2 b_2021-09-21.png (0211e954c556...), FinalLightCurve_WASP-2 b_2021-09-21.csv (e6d6ca9fd34e...)

Compute resources

Wall time 190.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 09:47Z.